University of Oxford, Hilary term 2027, Syllabus for:
Foundations of machine learning

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class time	Thursday 15:00-16:30 and Friday 11:30-13:00
webpage	https://maxkasy.github.io/home/ML_Oxford_2027/
canvas page	https://canvas.ox.ac.uk/courses/298679
location	Lecture Theatre, Manor Road Building

1 Overview and Objectives

The goal of this course is to provide you with solid **theoretical foundations in machine learning**. This will allow you to

1. become critical consumers of machine learning research, including an understanding of when new methods might or might not be useful for empirical work in economics,
2. develop your own research agenda around importing ideas from machine learning into economic and econometric theory, and
3. speak to the machine learning literature, contributing ideas from economics.

We begin by introducing some **foundations**. The class starts with a review of statistical decision theory, which provides the conceptual framework for the rest of the course. We then introduce basic ideas of supervised learning, using *Python* packages for machine learning. We next introduce probably approximately correct learning theory for classification and prediction. We then consider regularization and data-driven choice of tuning parameters. We will discuss the canonical normal means model. In this model, we will motivate shrinkage estimators in different ways, and will prove the famous result that shrinkage estimators can uniformly dominate conventional estimators.

We will then apply these general ideas and discuss several methods for **supervised machine learning**, that is, prediction. We will discuss Gaussian process regressions and deep neural nets. In this context, we will also consider numerical methods used for training neural nets, such as stochastic gradient descent. We finish this part of class by discussing conformal inference, a simple general technique to calculate valid confidence intervals for the predictions of any supervised learning algorithm.

The next part of class will cover **online learning**, where data arrive sequentially and decisions are made along the way, and **reinforcement learning**, where our actions additionally determine what we observe, and possibly the future states of

the world. We will start with online convex optimization, where no probabilistic assumptions about data generation are made at all, and we will explore the connection of statistical learning to optimization, including stochastic gradient descent. We will next consider multi-armed bandits, and review some theoretical results providing performance guarantees (regret bounds) for algorithms used for learning in bandit settings. We will then turn to a generalization of bandit problems, Markov decision problems, and will discuss reinforcement learning approaches for solving these, based on learning value functions, and on learning policies directly (policy gradient methods).

We will then turn to **large language models**, which bring together many of the tools from earlier parts of class. We will describe the training pipeline of a modern language model – pretraining, supervised fine-tuning, preference optimization, and reinforcement learning with verifiable rewards – as a sequence of decision problems, discuss the algorithms used to solve them, and introduce the transformer architecture.

The class will conclude with a discussion of **ethics** and the **social impact** of artificial intelligence. We will, in particular, review debates surrounding fairness and discrimination, as well as differential privacy.

If you need any special accommodations for the lectures, for physical or medical reasons, please send me an email.

2 Assignments

Your grade for this class will be based on four components. (The overall workload will be no greater than in previous years, but with more in-person components.)

1. A portfolio of 3 coding exercises, **due** on **6 April 2027** (each pass/fail; together 15% of grade).
2. A research proposal of 1200-1500 words, **due** on **6 April 2027** (25% of grade).
3. A 10 minute in-class presentation of your research proposal at the end of term, to take place in **Week 8** of HT (20% of grade).
4. A 2 hour exam after the end of term, which covers the theoretical background and interpretation for the coding exercises, to take place in **week 9** of HT (40% of grade).

Both the coding portfolio and the research proposal have to be submitted in **pdf** format via **Inspera**.

The four components are designed to reinforce each other. The **coding portfolio** is the preparation for the **exam**: the exam asks you to explain and interpret the methods and findings from the coding exercises. The **research proposal** is the preparation for the **presentation**: the presentation asks you to motivate and defend your proposed research in front of me and your classmates. Additionally, you will get to revise your proposal based on the feedback from the presentation. In both cases,

the take-home component is where you do the work, and the in-person component is where you demonstrate what you have understood.

Coding portfolio The portfolio of coding exercises will be based on three problem sets. The problem sets focus on implementing some of the methods discussed in class in **Python** or in **R**. Problem 1 of the first problem set and the second problem set use Python (with scikit-learn); for the other problems, the choice is yours. You will work with real data sets, and conduct simulation exercises to verify some of our theoretical results numerically. To help you learn these programming languages, if needed, I have compiled a list of resources here: `https://maxkasy.github.io/home/computationlinks/`.

I strongly encourage you to complete the problem sets early during the term, to avoid a time-crunch at the end of term, and to stay in sync with lectures. The suggested completion dates for the problem sets are 12 February, 26 February, and 12 March. I highly suggest finishing all three problem sets before the exam in week 9: the exam asks you to explain and interpret them, even though the portfolio is only due on 6 April. Your coding portfolio has to satisfy the following conditions:

- For each of the 3 assignments, all results should be generated from a single script, running from start to end, producing all the output.
- Output and discussion of findings have to be integrated in a report generated in R-Markdown or Quarto (for R), or a Jupyter notebook (for Python).
- Figures and tables have to be clearly labeled and interpretable.
- You have to submit the pdf (and only the pdf) generated from your script. This pdf should contain, first, a brief discussion and overview of your code. The pdf should then show the source code used to generate your findings. This pdf should, finally, show the figures and tables generated by your code, joint with a brief discussion of your findings.
- Please make this "human-readable" and do not include lists of warnings, intermediate calculations, etc.

You are allowed, and in fact encouraged, to collaborate in small groups (of up to 4 students) for the coding assignments. You are allowed to share jointly written computer code with the classmates in your group. However, each one of you must write up your answers completely independently.

Assessment of the coding portfolio. Each of the 3 assignments in the coding portfolio is assessed on a **pass/fail** basis, and together they count for 15% of your grade. All three exercises must be submitted and each must satisfy the conditions listed above: the script runs from start to finish and produces the required output, the figures and tables are labeled and interpretable, and the findings are discussed.

Your *understanding* of the coding exercises is assessed in the exam, which asks you to explain and interpret the methods and findings from the coding exercises. You may use AI assistance freely in producing the portfolio, but you will need to be able to account for what you submitted, without assistance, in the exam. A portfolio

that you did not genuinely engage with will leave you badly prepared for 40% of your grade.

Late submission. Please note that, under the MPhil Conventions, **late submission of a pass/fail element results in an instant fail of that element**. Submit on time.

Research proposal The research proposal has to be based on one of the areas of machine learning covered in class. Proposals might fall in one of several domains, including

(i) applications of machine learning in empirical economics,

(ii) econometric theory drawing on ideas from machine learning, and

(iii) machine learning theory, drawing on ideas from economics.

Your research proposals have to satisfy the following conditions:

1. They cannot be identical with the topic of the MPhil thesis that you might be writing concurrently.
2. The proposals have to indicate clearly whether your contribution is of type (i), (ii), or (iii), as described above.
3. The proposals should be in 12pt font, 1.5 linespacing, margins of 3cm, A4. The length should be 1200-1500 words. Code and references do not count to this word limit.
4. The proposals cannot consist exclusively of a summary of literature; they have to propose new research.

The proposal is marked on the substance of the proposed research: The originality and interest of the idea, the clarity of the contribution, and the feasibility of the approach. It is also your preparation for the presentation, where you will be asked to motivate and defend the proposal in person.

Presentation In the last week of class, you will give a presentation of your research proposal.

- The proposal does not need to be finished at this point. The presentation is an opportunity to obtain feedback from me and your classmates.
- The presentation should be no longer than 10 minutes, and will be followed by questions from me and from your classmates.
- The presentation is assessed on the clarity of the presentation itself, and on your ability to motivate, situate, and defend your proposed research in the discussion that follows.
- I recommend no more than 5 slides, at most 2 formulas per slide, and no more than 10 numbers per slide (if you have simulations or empirical results).

Exam The 2 hour exam will focus on the theoretical background for the three coding exercises, and on the interpretation of the results that you produced in them.

- The exam will ask you to explain the methods you implemented, to interpret the output of your simulations in light of the theoretical results derived in class, and to reason about how your results would change under modified assumptions.
- Additionally, you will be asked to explain pieces of code from my solution of the exercises.
- The exam will be computer-based and will take place in the exam schools.
- More detail will be provided in class towards the end of term.

Policy on the use of generative AI The assessment for this course is designed on the assumption that you have access to generative AI, and that you will use it. You are free to use it for the coding portfolio and the research proposal. The exam and the presentation are the components where you demonstrate, in person and unassisted, that you understand what you produced. The take-home components are, accordingly, the means by which you prepare for the in-person ones. Using AI to produce work you do not understand is allowed but would not be helpful to you.

The department policy on generative AI applies: *When undertaking study and research for an MPhil course or thesis you are permitted to make selective use of AI in your work for tasks such as the following:*

- *Brainstorming ideas*
- *Surveying literature relevant to your area of study*
- *Gathering data and generating code to implement estimation techniques in econometric work*
- *Generating diagrams, charts and results tables for inclusion in submissions and/or the thesis*
- *Proof-reading and obtaining feedback on text that you have written*

It is important that you use AI as an enabler of your work rather than relying on AI to generate the intellectual value added in your submitted work/ thesis; in particular, the decisions that you take on how to implement research methods and the conclusions that you draw from them should be based on your own reasoning.

In order to promote fair use of AI the University provides all students with access to tools developed by OpenAI. Before making use of these tools please read the University guidelines on the use of AI available at the following link: `https://www.ox.ac.uk/students/academic/guidance/skills/ai-study`

When students submit an assignment or thesis on Inspera they are asked to complete a Declaration of Authorship. For Economics submissions this will include a question that asks candidates to provide a clear statement of the different tasks that they have completed with the assistance of AI. The information provided by candidates will be shared with examiners and thesis assessors.

3 References

Required readings are marked by ☺. Topics marked by † are not covered in lectures; for these, bonus slides are available on the course website.

3.1 Foundations

Review of decision theory

☺Robert, C. (2007). *The Bayesian choice: from decision-theoretic foundations to computational implementation*. Springer Verlag, chapter 2.

Machine learning using Python

VanderPlas, J. (2022). *Python Data Science Handbook*. O'Reilly, chapter 5.

Probably approximately correct learning theory

☺Shalev-Shwartz, S. and Ben-David, S. (2014). *Understanding machine learning: From theory to algorithms*. Cambridge University Press, chapters 2 to 6.

Shrinkage in the normal means model

Wasserman, L. (2006). *All of nonparametric statistics*. Springer Science & Business Media, chapter 7.

☺Stigler, S. M. (1990). The 1988 Neyman memorial lecture: a Galtonian perspective on shrinkage estimators. *Statistical Science*, pages 147–155.

☺Morris, C. N. (1983). Parametric empirical Bayes inference: Theory and applications. *Journal of the American Statistical Association*, 78(381):pp. 47–55.

Stein, C. M. (1981). Estimation of the mean of a multivariate normal distribution. *The Annals of Statistics*, 9(6):1135–1151.

Abadie, A. and Kasy, M. (2019). Choosing among regularized estimators in empirical economics - the risk of machine learning. *Review of Economics and Statistics*, 101(5).

Fessler, P. and Kasy, M. (2019). How to Use Economic Theory to Improve Estimators: Shrinking Toward Theoretical Restrictions. *The Review of Economics and Statistics*, 101(4):681–698.

Limiting experiments and the normal means model†

van der Vaart, A. W. (2000). *Asymptotic statistics*. Cambridge University Press, chapter 7.

Hansen, B. E. (2016). Efficient shrinkage in parametric models. *Journal of Econometrics*, 190(1):115–132.

3.2 Supervised learning

Gaussian process priors

☺Williams, C. and Rasmussen, C. (2006). *Gaussian processes for machine learning*. MIT Press, chapters 2 and 7.

Reproducing kernel Hilbert spaces and splines†

Wahba, G. (1990). *Spline models for observational data*, volume 59. Society for Industrial Mathematics, chapter 1.

Applications of Gaussian process priors in economics†

Kasy, M. (2016). Why experimenters might not always want to randomize, and what they could do instead. *Political Analysis*, 24(3):324–338.

Kasy, M. (2018). Optimal taxation and insurance using machine learning – sufficient statistics and beyond. *Journal of Public Economics*, 167.

Regression trees and random forests†

Friedman, J., Hastie, T., and Tibshirani, R. (2001). *The elements of statistical learning*, volume 1. Springer series in statistics Springer, Berlin, chapters 8 and 9.

Athey, S. and Imbens, G. (2016). Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences*, 113(27):7353–7360.

Deep neural nets

☺Goodfellow, I., Bengio, Y., and Courville, A. (2016). *Deep learning*. MIT Press, chapters 6-8.

Bottou, L., Curtis, F. E., and Nocedal, J. (2018). Optimization methods for large-scale machine learning. *SIAM Review*, 60(2):223–311

Bartlett, P. L., Montanari, A., and Rakhlin, A. (2021). Deep learning: a statistical viewpoint. *arXiv preprint arXiv:2103.09177*

Image generation, autoencoders, and diffusion models†

Luo, C. (2022). Understanding diffusion models: A unified perspective. *arXiv preprint arXiv:2208.11970*.

Yang, L., Zhang, Z., Song, Y., Hong, S., Xu, R., Zhao, Y., Zhang, W., Cui, B., and Yang, M.-H. (2023). Diffusion models: A comprehensive survey of methods and applications. *ACM Computing Surveys*, 56(4):1–39.

Conformal inference

☺Angelopoulos, A. N. and Bates, S. (2021). A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*

Double/debiased machine learning†

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.

3.3 Online learning and reinforcement learning

Online convex optimization

☺Hazan, E. (2016). Introduction to online convex optimization. *Foundations and Trends in Optimization*, 2(3-4):157–325, chapter 3.

Adversarial online learning†

Cesa-Bianchi, N. and Lugosi, G. (2006). *Prediction, learning, and games*. Cambridge University Press, chapter 2.

Bandit problems

Bubeck, S. and Cesa-Bianchi, N. (2012). Regret Analysis of Stochastic and Nonstochastic Multi-armed Bandit Problems. *Foundations and Trends® in Machine Learning*, 5(1):1–122.

☺Russo, D. J., Roy, B. V., Kazerouni, A., Osband, I., and Wen, Z. (2018). A Tutorial on Thompson Sampling. *Foundations and Trends® in Machine Learning*, 11(1):1–96.

Caria, S., Gordon, G., Kasy, M., Osman, S., Quinn, S., and Teytelboym, A. (2024). Job search assistance for refugees in Jordan: An adaptive field experiment. *Journal of the European Economic Association*, 22(2).

Weber, R. et al. (1992). On the Gittins index for multiarmed bandits. *The Annals of Applied Probability*, 2(4):1024–1033.

Wager, S. and Xu, K. (2021). Diffusion asymptotics for sequential experiments. *arXiv preprint arXiv:2101.09855*.

Cesa-Bianchi, N., Colomboni, R., and Kasy, M. (2025). Adaptive maximization of social welfare. *Econometrica*, 93(3).

Lewi, Y., Kaplan, H., and Mansour, Y. (2020). Thompson sampling for adversarial bit prediction. In *Algorithmic Learning Theory*, pages 518–553. PMLR.

Experiments for policy choice[†]

Kasy, M. and Sautmann, A. (2021). Adaptive treatment assignment in experiments for policy choice. *Econometrica*, 89(1):113–132.

Russo, D. (2016). Simple bayesian algorithms for best arm identification. In *Conference on Learning Theory*, pages 1417–1418.

Markov decision problems and reinforcement learning

☺Sutton, R. S. and Barto, A. G. (2018). *Reinforcement learning: An introduction*. MIT press, chapters 3, 4, 6, and 13.

François-Lavet, V., Henderson, P., Islam, R., Bellemare, M. G., and Pineau, J. (2018). An introduction to deep reinforcement learning. *Foundations and Trends® in Machine Learning*, 11(3-4):219–354.

☺Agarwal, A., Brantley, K., Kakade, S. M., Jiang, N., and Sun, W. (2026). Reinforcement learning: Theory and algorithms. `https://rltheorybook.github.io/`. Working draft, chapter 9.

3.4 Large language models

☺Jurafsky, D. and Martin, J. H. (2026). Speech and Language Processing. 3rd edition draft, `https://web.stanford.edu/~jurafsky/slp3/`, chapter 7.

Phuong, M. and Hutter, M. (2022). Formal algorithms for transformers. *arXiv preprint arXiv:2207.09238*.

Lambert, N. (2026). Reinforcement Learning from Human Feedback. `https://rlhfbook.com/`, chapters 4–8 and 14.

Rafailov, R., Sharma, A., Mitchell, E., Manning, C. D., Ermon, S., and Finn, C. (2023). Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36.

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. (2022). Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35.

Team Olmo (2025). Olmo 3. *arXiv preprint arXiv:2512.13961*.

3.5 Ethics and machine learning

☺Kearns, M. and Roth, A. (2019). *The Ethical Algorithm: The Science of Socially Aware Algorithm Design*. Oxford University Press.

☺Kasy, M. (2025). *The means of prediction: How AI Really Works (and Who Benefits)*. University Of Chicago Press.

Fairness

Pessach, D. and Shmueli, E. (2020). Algorithmic fairness. *arXiv preprint arXiv:2001.09784*

Kasy, M. and Abebe, R. (2021). Fairness, equality, and power in algorithmic decision making. *ACM Conference on Fairness, Accountability, and Transparency*.

Kasy, M. (2024). Algorithmic bias and racial inequality: A critical review. *Oxford Review of Economic Policy*.

Roemer, J. E. (1998). *Theories of distributive justice*. Harvard University Press, Cambridge.

Differential privacy

Dwork, C. and Roth, A. (2014). The algorithmic foundations of differential privacy. *Foundations and Trends® in Theoretical Computer Science*, 9(3–4):211–407.

Explainability[†]

Lipton, Z. C. (2018). The mythos of model interpretability: In machine learning, the concept of interpretability is both important and slippery. *Queue*, 16(3):31–57.

Wachter, S., Mittelstadt, B., and Russell, C. (2018). Counterfactual explanations without opening the black box: automated decisions and the GDPR. *Harvard Journal of Law and Technology*, 31(2).